MCTA 3203

MECHATRONICS SYSTEM INTEGRATION

SEMESTER 1, 25/26

WEEK 08 : BLUETOOTH DATA INTERFACING

SECTION 2

GROUP 19

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ABSTRACT

This experiment focuses on the development of a wireless temperature monitoring and control system using Bluetooth communication. An Arduino Mega integrated with a DHT11 temperature and humidity sensor and an HC-05/HC-06 Bluetooth module was employed to transmit real-time environmental data to a smartphone or computer. Additionally, control commands sent wirelessly via Bluetooth were used to toggle an external LED, simulating a fan control system. The experiment demonstrates fundamental concepts of sensor interfacing, Bluetooth serial communication, and remote monitoring in mechatronics system integration. The results confirm reliable data transmission and effective remote control, highlighting Bluetooth as a suitable medium for short-range wireless monitoring applications.

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1.0 INTRODUCTION

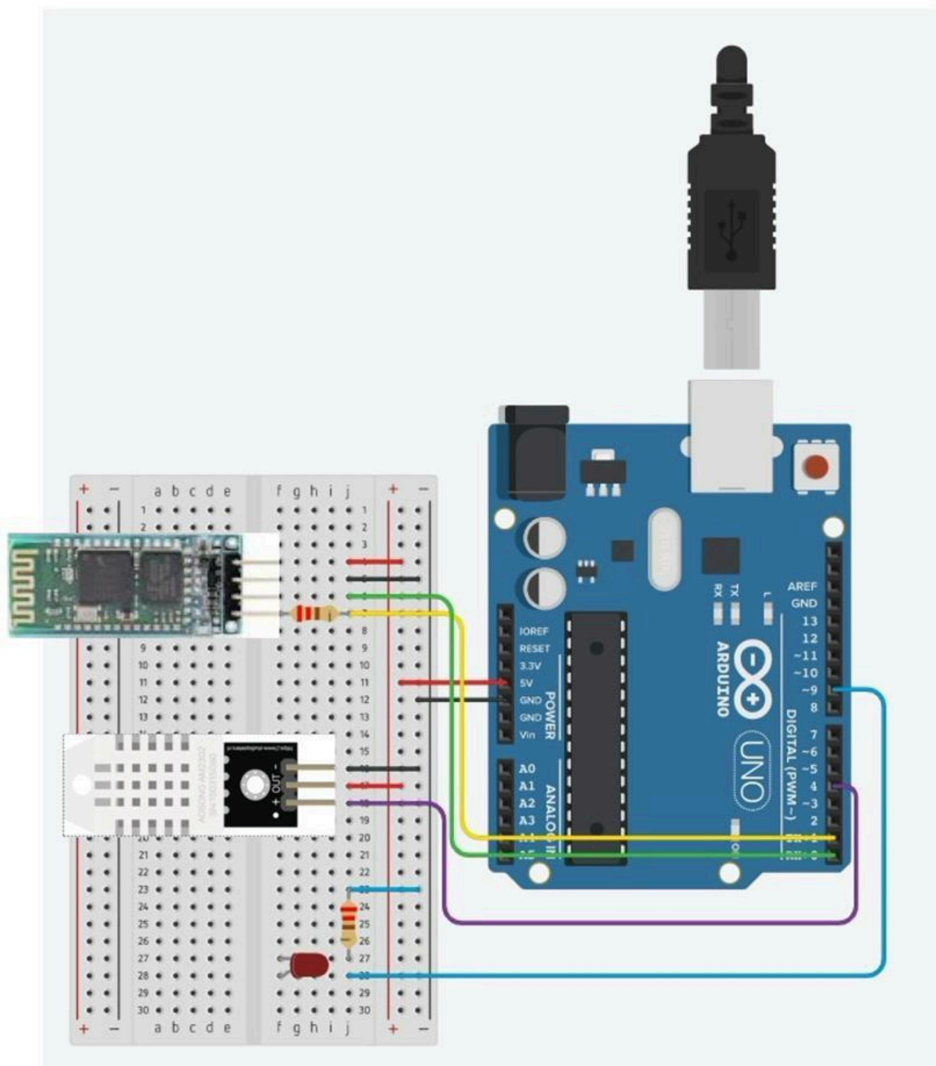
In this experiment, we created a simple wireless monitoring system that allows temperature and humidity readings to be viewed from a smartphone or computer using Bluetooth. The system uses an Arduino, a DHT11 sensor, and an HC-06 Bluetooth module. The Arduino reads the temperature and humidity from the sensor, then sends the data wirelessly to a paired device. At the same time, the user can send commands such as “FAN ON” or “FAN OFF” to control an LED, which represents a fan.

Through this experiment, we learned how sensors, Bluetooth communication, and simple control actions can work together to form a basic IoT-style system. This setup reflects how real-world devices, such as smart home appliances, communicate and respond to user inputs.

2.0 MATERIALS AND EQUIPMENT

Item	**Quantity**	**Description**
Laptop	1	Running Arduino
Smartphone	1	To control the LED on/off
Arduino Board	1	Arduino Mega 2560
Resistors	2	220 Ω
LED	1	Represent fan
Breadboard	1	Circuit mounting
Jumper wires	As needed	Signal connections
DHT11	1	To read temperature and humadity
HC-06	1	Bluetooth module

3.0 EXPERIMENTAL SETUP



4.0 METHODOLOGY

1. The Arduino Mega was initialized using the Arduino IDE, and the required libraries for the DHT11 temperature sensor were included in the program.
2. The hardware components (DHT11 sensor, HC-06 Bluetooth module, and external LED) were connected to the Arduino Mega according to the defined pin configuration.
3. The HC-06 Bluetooth module was paired with a smartphone using the default Bluetooth pairing procedure.
4. An Arduino Bluetooth Control application was installed on the smartphone to enable wireless data monitoring and control.
5. The Arduino program was written to:
   - Read temperature and humidity values from the DHT11 sensor.
   - Transmit the sensor data to the smartphone via Bluetooth.
   - Receive user commands ("FAN ON" and "FAN OFF") from the smartphone.
   - Control the external LED based on the received commands.

6. The Arduino code was compiled and uploaded to the Arduino Mega using the Arduino IDE.
7. Once powered, the system continuously measured the temperature and transmitted the readings to the smartphone at fixed time intervals.
8. The user sent control commands from the smartphone to the Arduino Mega via Bluetooth to switch the LED ON and OFF remotely.
9. The output response (LED ON/OFF) was observed and verified in real time to confirm correct wireless control operation.
10. The temperature data received was observed on the smartphone and could optionally be recorded on a computer for further analysis.

Code Used :

```
#include "DHT.h"

// --------------------------
// HARDWARE CONFIG
// --------------------------
#define DHTPIN 4        // DHT11 data pin
#define DHTTYPE DHT11
#define LEDPIN 8        // External LED / FAN indicator

DHT dht(DHTPIN, DHTTYPE);

// --------------------------
// SETUP
// --------------------------
void setup() {
  pinMode(LEDPIN, OUTPUT);
  digitalWrite(LEDPIN, LOW);  // LED starts OFF

  Serial.begin(9600);   // USB Serial for debug
  Serial1.begin(9600);  // Hardware Serial1 for HC-06

  dht.begin();

  Serial.println("Mega DHT11 + Bluetooth LED Control Ready");
  Serial1.println("Mega DHT11 + Bluetooth LED Control Ready");
}

// --------------------------
// MAIN LOOP
// --------------------------
void loop() {
  // ----- READ SENSOR -----
  float temp = dht.readTemperature(); // Celsius
  float hum  = dht.readHumidity();

  // Send sensor data via Bluetooth AND USB Serial Monitor
  if (!isnan(temp) && !isnan(hum)) {
    String tempStr = "TEMP:" + String(temp, 1);
    String humStr  = "HUM:"  + String(hum, 1);

    Serial.println(tempStr);
    Serial.println(humStr);

```

```
    Serial1.println(tempStr);
    Serial1.println(humStr);
  } else {
    Serial.println("Failed to read from DHT11!");
    Serial1.println("Failed to read from DHT11!");
  }

  // ----- RECEIVE BLUETOOTH COMMANDS -----
  if (Serial1.available()) {
    String cmd = Serial1.readStringUntil('\n'); // Read one line
    cmd.trim();  // Remove whitespace

    String fanStatus;

    if (cmd.equalsIgnoreCase("FAN ON")) {
      digitalWrite(LEDPIN, HIGH); // Turn LED ON
      fanStatus = "FAN:ON";
    }
    else if (cmd.equalsIgnoreCase("FAN OFF")) {
      digitalWrite(LEDPIN, LOW);  // Turn LED OFF
      fanStatus = "FAN:OFF";
    }
    else {
      fanStatus = "FAN:UNKNOWN";
    }

    // Send fan status updates to both ports
    Serial1.println(fanStatus);
    Serial.println(fanStatus);
  }

  delay(2000); // 2 seconds between readings
}
```

5.0 DATA COLLECTION

The DHT11 temperature sensor continuously measured the ambient temperature and humidity at 2-second intervals throughout the experiment. The measured values were displayed on the Serial Monitor of the computer and simultaneously transmitted wirelessly to the smartphone through the HC-06 Bluetooth module. The temperature readings were observed in real time using the Arduino Bluetooth Control application on the smartphone. Multiple temperature readings were manually recorded at regular time intervals over a fixed duration. At the same time, the ON and OFF status of the LED, which simulated a fan, was observed and recorded whenever control commands (“FAN ON” or “FAN OFF”) were sent from the smartphone. Any noticeable changes in temperature when the LED was turned ON or OFF were also noted for further analysis.

6.0 DATA ANALYSIS

The temperature and humidity data transmitted by the DHT11 sensor through the Bluetooth module were examined to understand how accurately and consistently the system measured environmental conditions. The recorded temperature readings were reviewed over time to observe how the values changed under normal room conditions. By analyzing the readings at regular intervals, the experimenters were able to identify patterns such as steady temperature behavior or gradual increases due to surrounding heat. The ON and OFF status of the LED, which simulated a fan, were also compared with the Bluetooth commands sent from the smartphone to ensure proper system response. Any sudden fluctuations in temperature readings or delays in LED response were analyzed to determine whether they were caused by sensor limitations, Bluetooth signal interference, or power stability. Overall, the analysis helped evaluate the accuracy of the temperature monitoring system and the reliability of the wireless control during the experiment.

7.0 RESULT

The Bluetooth temperature monitoring system functioned as intended. The DHT11 sensor successfully measured environmental data, which was transmitted wirelessly to a paired device. Control commands sent through Bluetooth were correctly interpreted by the Arduino, resulting in accurate LED control. The experiment achieved real-time monitoring and remote control functionality**.**

8.0 DISCUSSION

From the program used, the system was able to successfully collect and transmit sensor data in real time. Every two seconds, the Arduino read the temperature and humidity values from the DHT11 sensor. If the readings were valid, the values were sent to both the Serial Monitor and the Bluetooth-connected device. This made it easy for us to monitor the environment from two different platforms at the same time.

The Bluetooth communication worked smoothly. Whenever a user sent a command either "FAN ON" or "FAN OFF" the Arduino could recognise the command, change the LED state, and send back a confirmation message. This shows how Bluetooth can be used not only for sending data, but also for receiving instructions and controlling physical hardware. Overall, the experiment helped us understand how a simple control system operates: it reads inputs, processes them, and reacts through outputs.

The entire setup demonstrated how microcontroller systems can be used for remote monitoring and control. Even though the system is simple, it can be further expanded into practical applications such as home automation, temperature-controlled ventilation systems, and wireless sensor networks. The experiment also highlighted the importance of stable communication and proper coding to ensure smooth data flow between devices.

9.0 CONCLUSION

In conclusion, a functional Bluetooth-based remote temperature monitoring and control system was successfully developed. The system was able to transmit real-time environmental data and respond to control commands wirelessly. This experiment enhances understanding of Bluetooth interfacing, serial communication, and remote system control, which are fundamental concepts in mechatronics system integration.

10.0 RECOMMENDATIONS

Although the system worked well, there are several improvements that could make it more efficient and practical. One of the main upgrades would be to use a more accurate sensor such as the DHT22 or BME280, as the DHT11 has limited accuracy and reacts slowly to temperature changes. The system can also be enhanced by adding automatic fan control, where the Arduino turns the fan on or off based on temperature thresholds instead of relying only on manual commands. This would make the system smarter and more responsive.

Additionally, creating a simple mobile app for monitoring and control would provide a better user experience compared to using a Bluetooth terminal. With a dedicated interface, users could view real-time graphs, press buttons to control the fan, and receive notifications if the temperature becomes too high. It would also be useful to implement a data logging feature so that temperature trends can be analysed over longer periods. Lastly, upgrading to an ESP32 board would allow the system to use both Bluetooth and Wi-Fi, making it possible to access the sensor data from anywhere, not just within Bluetooth range. These improvements would help transform the basic experiment into a more reliable and versatile IoT system.

11.0 REFERENCES

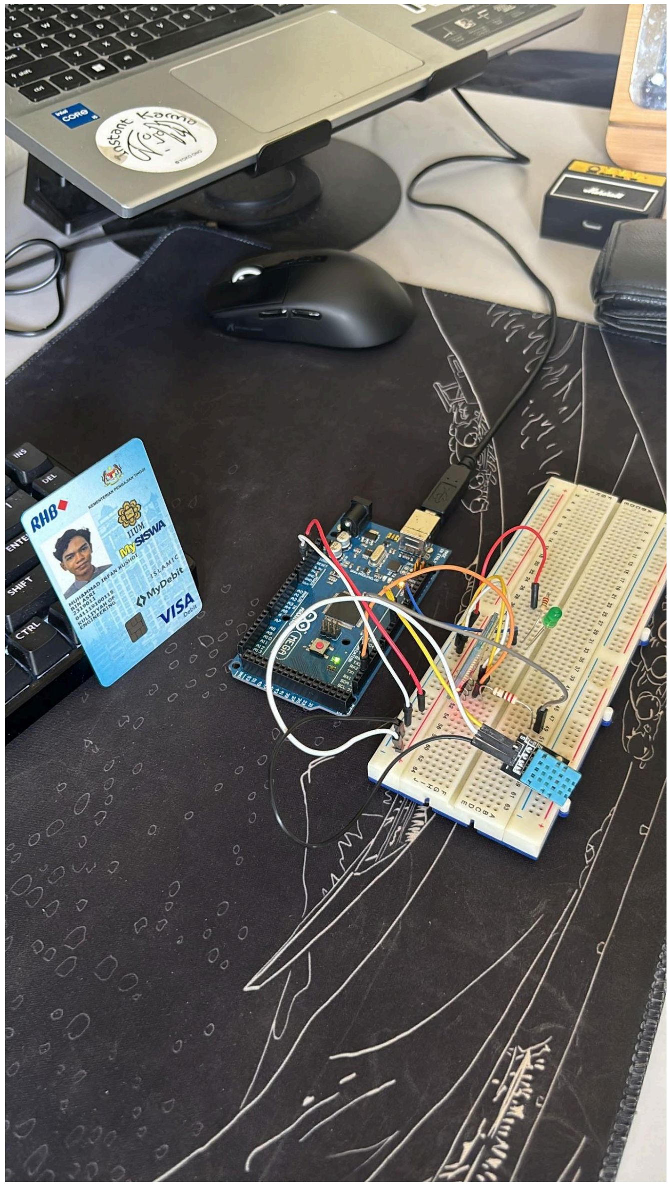
1.https://projecthub.arduino.cc/NeilChaudhary/arduino-bluetooth-basic-tutorial-9cff12Arduino Bluetooth Basic Tutorial

2.https://howtomechatronics.com/tutorials/arduino/arduino-and-hc-05-bluetooth-moduletutorial/ Arduino and HC-05 Bluetooth Module Complete Tutorial

3.https://projecthub.arduino.cc/superturis/basic-bluetooth-communication-with-arduino-hc-05-3a431c

Basic Bluetooth communicaƟon with Arduino & HC-05

APPENDICES



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Special thanks to DR WAHJU SEDIONO for their guidance and support during this experiment .

STUDENTS'S DECLARATION

CERTIFICATE OF ORIGINALITY AND AUTHENTICITY

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specific in references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or person.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report.** The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of individual's contributor to the report.

We, therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** have been **verified by us.**

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Contribution:	Introduction , Discussion , Recommendation		

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Contribution:	Abstract, result, conclusion, Reference		